

REPORT

ARTIFICIAL INTELLIGENCE

Assessment – 2: Constraint-Based Problem Solving

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1. INTRODUCTION

Constraint-Based Problem Solving is an important area of Artificial Intelligence that focuses on finding solutions while satisfying a set of predefined constraints. These problems are commonly solved using search algorithms and intelligent decision-making techniques. Constraint satisfaction is widely used in scheduling, path planning, robotics, resource allocation, and optimization problems. This assessment includes three AI problems: Hospital Doctor Shift Assignment, Robot Grid Navigation, and Autonomous Rescue Robot Navigation. Each problem demonstrates the application of search algorithms and intelligent agents to obtain feasible and optimal solutions under different constraints.

2. OBJECTIVE

The main objectives of this assessment are:

- To understand the concept of Constraint-Based Problem Solving.
- To analyze different types of constraints in AI problems.
- To apply Backtracking Search for scheduling problems.
- To implement Breadth-First Search (BFS) for pathfinding.
- To understand Uniform Cost Search and Online Search Agents.
- To improve logical reasoning and problem-solving skills using Artificial Intelligence.

3. SOFTWARE REQUIREMENTS

- Python 3.x
- Visual Studio Code / Notepad
- GitHub

- Windows Operating System

4. PROBLEMS IMPLEMENTED

Problem 1 – Hospital Doctor Shift Assignment

The Hospital Doctor Shift Assignment problem is a Constraint Satisfaction Problem (CSP) in which three doctors (D1, D2, and D3) are assigned to Morning, Afternoon, and Night shifts while satisfying all scheduling constraints. The Backtracking Search algorithm systematically checks every possible assignment and backtracks whenever a constraint is violated. The final schedule ensures that each doctor is assigned exactly one shift without conflicts.

Constraints

- D1 cannot work the Night shift.
- D2 must work before D3.
- Only one doctor is assigned to each shift.
- D3 cannot work the Morning shift.
- Every doctor is assigned exactly one shift.

Problem 2 – Robot Grid Navigation

The Robot Grid Navigation problem represents a robot moving in a 5×5 grid from the Start position to the Goal while avoiding obstacles. The robot is allowed to move only in four directions, and each movement has a cost of one unit. Breadth-First Search (BFS) explores all possible paths level by level and guarantees the shortest path to the destination while satisfying the movement constraints.

Constraints

- Move Up, Down, Left, and Right.
- Movement cost equals one.
- Obstacles cannot be crossed.
- Manhattan Distance is used as the heuristic.

Problem 3 – Autonomous Rescue Robot

The Autonomous Rescue Robot operates in a disaster-affected building represented as a grid. The robot must rescue survivors while avoiding blocked paths and risky zones. Since the environment is partially observable, the robot updates its knowledge dynamically while navigating. Uniform Cost Search (UCS) is used to identify the least-cost path by considering both movement costs and additional costs for risky areas.

Constraints

- Movement in four directions only.
- Blocked cells cannot be traversed.

- Limited sensing capability.
- Movement cost equals one.
- Risky zones add an extra cost.
- Dynamic knowledge updating.
- Minimize the total navigation cost.

5. ALGORITHMS USED

- Backtracking Search Algorithm
- Breadth-First Search (BFS)
- Uniform Cost Search (UCS)
- Online Search Agent

6. AI CONCEPTS APPLIED

The implementation demonstrates several important Artificial Intelligence concepts:

- Constraint Satisfaction Problems (CSP)
- Intelligent Agents
- Rational Decision Making
- State-Space Search
- Backtracking Search
- Breadth-First Search
- Uniform Cost Search
- Online Search
- Path Planning
- Cost Optimization

7. APPLICATIONS

The concepts implemented in this assessment have several real-world applications:

- Hospital staff scheduling
- Employee shift management
- Robot navigation systems
- Autonomous rescue operations
- Disaster management
- Warehouse automation
- Route optimization
- Smart transportation systems
- Industrial robotics
- Artificial Intelligence planning systems

8. ADVANTAGES

- Improves logical reasoning and analytical thinking.
- Demonstrates practical AI search algorithms.

- Provides knowledge of constraint satisfaction problems.
- Improves optimization and decision-making skills.
- Helps understand intelligent agent behavior.
- Enhances programming and problem-solving abilities.

9. RESULT

All three constraint-based problems were successfully analyzed and solved using appropriate Artificial Intelligence techniques. Backtracking Search generated a valid doctor scheduling solution, Breadth-First Search identified the shortest path for robot navigation, and Uniform Cost Search produced the least-cost path for the autonomous rescue robot. The objectives of the assessment were successfully achieved.

10. CONCLUSION

This assessment provided practical knowledge of Constraint-Based Problem Solving in Artificial Intelligence. The Hospital Doctor Shift Assignment demonstrated the use of Backtracking Search for solving scheduling problems. The Robot Grid Navigation problem illustrated how Breadth-First Search efficiently finds the shortest path in a grid environment. The Autonomous Rescue Robot demonstrated the application of Uniform Cost Search and intelligent agents in dynamic environments. Overall, the assessment strengthened the understanding of AI search strategies, constraint satisfaction, optimization, and intelligent decision-making techniques for solving real-world problems.